

Claims Copilot

Intelligent Workflow Platform for Supplemental Health Claims

Prepared for: Prudential Financial Version: Prototype v2 Classification: Confidential

Executive Summary

Claims Copilot is an AI-powered workflow platform designed for supplemental health claims examiners. It consolidates eligibility verification, fraud detection, policy intelligence, and workflow management into a single interface — replacing the need to toggle between multiple systems for every claim.

The platform scores and ranks all incoming claims using a three-layer intelligence pipeline, surfaces fraud patterns that would be invisible when reviewing claims individually, and provides an AI copilot that assists the examiner through every step of the investigation process.

The core value proposition is simple: examiners process claims faster, catch more fraud, and document everything automatically.

What the Platform Does

For Every Claim (Daily Workflow Value)

Today (Without Platform)	With Claims Copilot
Check eligibility across 3 separate systems	One-click eligibility check returns a consolidated result
Manually cross-reference policy terms, state rules, and ET resolution	Copilot surfaces applicable rules, exclusions, and coverage determination automatically
Track PMR, CBR, and TAT deadlines across different screens	Workflow tasks appear alongside the claim in a single view
Manually check if policy changes affect the claim	Proactive alerts surface relevant policy changes (e.g., PA-001 surgical repair update)
Write up case notes and escalation packages from scratch	Dossier generated automatically with timeline, evidence, and recommended actions

For Flagged Claims (Fraud Intelligence)

Capability	How It Works
Risk Scoring	Every claim receives a 0–100 risk score based on 30+ weighted features across policy, member, claim, provider, network, and temporal dimensions
Rules Engine	14 deterministic rules run on every claim (e.g., dependent spike, claim velocity, policy manipulation, duplicate submissions)
Document Analysis	13 automated checks on medical record metadata — font consistency, erasure indicators, missing headers, editable formats, and more
Network Detection	Entity graph connects members, dependents, providers, addresses, and employers to reveal rings, clusters, and coordinated patterns
Investigation Checklist	Structured 7-step process that can run automatically or step-by-step, with pass/fail tracking

Platform Views

1. Claims Queue — The Morning View

The examiner’s landing page. All 500 claims displayed in a sortable, filterable queue.

CLAIMS QUEUE [Filter ▼] [🔍]					
[All (500)] [Wellness (250)] [Accident (100)] [HI (100)] [CI (50)]					
Risk: [🔴 HIGH (12)] [🟡 MEDIUM (38)] [🟢 LOW (450)]					
Claim#	Type	Member	Flag / Task	Risk	Status
WC-247	Wellness	M. Rivera	▶ 35 deps added	🔴 94	NEW
CI-051	Crit Ill	R. Torres	▶ Owner change 45d	🔴 88	NEW
AC-103	Accident	J. Park	▶ Tampered records	🟡 76	NEW
HI-089	Hosp Ind	T. Chen	📄 PMR open (2d)	🟡 65	REVIEW
AC-107	Accident	D. Williams	📞 CBR pending	🟢 28	NEW
WL-401	Wellness	S. Johnson	✓ Auto-adjudicated	🟢 5	APPROVED

Key design principle: Fraud flags and workflow tasks (PMR, CBR, TAT) appear side by side. The examiner sees their full workload — not just the suspicious claims.

Claims are filterable by:

- **Claim type:** Wellness, Accident, Hospital Indemnity, Critical Illness
- **Risk tier:** High (65–100), Medium (30–64), Low (0–29)
- **Status:** New, In Review, Escalated, Approved, Denied, Closed
- **Workflow state:** Open PMR, Pending CBR, Past TAT

2. Copilot — AI-Assisted Claim Review

Selecting any claim opens the Copilot — a conversational AI assistant with full access to the claim's data, related entities, and investigation tools.

CLAIM WC-247 — M. RiveraRisk: ● 94

Wellness | Acme Corp | Coverage ends: Sept 2025

⚠️ 3 rules triggered | 📄 PMR open | 🚫 No active CBR

Checklist Progress

✓ Initial

✓ Eligible

● Fraud

■ Family

■ Medical

■ Policy

■ Determine

Examiner: Why was this flagged?

Copilot: This claim was flagged for three reasons:

1. Dependent Spike (R-001, BLOCK): M. Rivera added 35 dependents in 14 days. Threshold is 5 in 30 days.

2. Claim Velocity (R-002, FLAG): 47 claims this quarter across member and dependents. Threshold is 10.

3. Benefit Max Gaming (R-009, INFO): All 35 wellness claims filed at exactly \$100.00 — the benefit max.

Risk score breakdown: policy age (low risk), member features (very high — dependent count is 8σ above mean), network signals (shared address with 2 other members).

Shall I run the full investigation checklist, or would you like to start with the dependent analysis?

[Type your question...]

[Send]

[▶ Run Full Checklist]

[Dependent Check]

[Claim History]

[Document Analysis]

[View Network]

[Escalate]

The Copilot can:

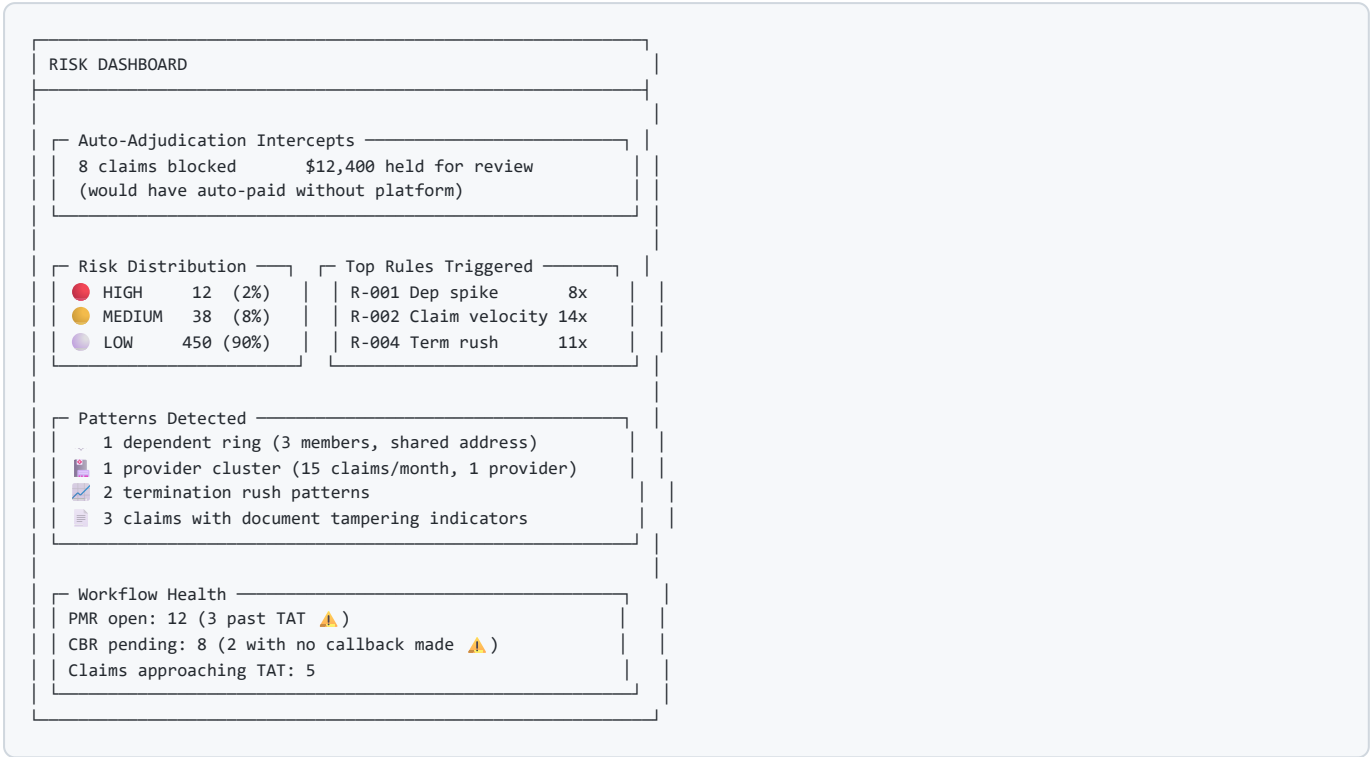
- Answer natural language questions about any aspect of the claim
- Call investigation tools on demand (eligibility, documents, family claims, provider patterns)
- Run the 7-step checklist automatically or one step at a time
- Proactively surface relevant policy alerts and workflow tasks
- Present red flags alongside innocent explanations — it doesn't just accuse
- Generate escalation packages for SIU referral

Quick action chips adapt to the claim type:

- **Wellness:** Dependent Check, Claim History, Auto-Adjudication Review
- **Accident:** Match to Policy, Accident Details, State Rules
- **Hospital Indemnity:** Admission/Discharge, Provider Check, Policy Terms
- **Critical Illness:** Medical Records, Document Analysis, Policy Check

3. Risk Dashboard — The Morning Briefing

Aggregate intelligence view showing the examiner what matters most today.



Auto-Adjudication Intercepts shows the direct financial impact — claims that would have auto-paid without the platform’s intervention.

Patterns Detected surfaces network-level intelligence that is invisible when reviewing claims individually. Each pattern is clickable, navigating directly to the relevant claims.

Workflow Health ensures TAT compliance and follow-up accountability.

4. Network Graph — Relationship Visualization

Interactive entity graph showing connections between members, dependents, providers, addresses, and employers.



The graph reveals:

- **Dependent rings:** Unrelated members sharing the same dependents
- **Address clusters:** Multiple claimants at the same physical address
- **Provider mills:** Abnormal patient volume through a single provider
- **Employer-linked patterns:** Claim spikes from a single employer group

Three-Layer Intelligence Pipeline

Every claim passes through three layers of analysis before an examiner sees it.

Layer 1: Rules Engine (14 Deterministic Rules)

Fast, explainable business rules that fire on every claim. Each rule has a severity level:

Severity	Meaning	Example Rules
BLOCK	Prevents auto-adjudication, requires examiner review	Dependent Spike (>5 in 30 days), Duplicate Submission
FLAG	Surfaces in queue for investigation	Claim Velocity, Termination Rush, Policy Manipulation, Provider Cluster, Resubmission After Denial
INFO	Noted in checklist, doesn't force review	Benefit Max Gaming, CBR Frequency, Medical Records Via Email, Policy Change Impact

Full rule set:

ID	Rule	Condition	Severity
R-001	Dependent Spike	>5 dependents added in 30 days	BLOCK
R-002	Claim Velocity	>10 claims per member per quarter	FLAG
R-003	Early Filing	Claim <90 days after policy start	FLAG
R-004	Termination Rush	Claim spike within 60 days of coverage end	FLAG
R-005	Provider Cluster	Same provider + same diagnosis + 5 patients in 30 days	FLAG
R-006	Policy Manipulation	Owner/beneficiary change <180 days before claim	FLAG
R-007	Dependent Age Gap	Member-dependent age gap >25 AND dependent age >25	FLAG
R-008	Duplicate Submission	Same member, date of service, and amount	BLOCK
R-009	Benefit Max Gaming	3+ consecutive claims at exact benefit maximum	INFO
R-010	Resubmission After Denial	Denied claim resubmitted with <3 field changes	FLAG
R-011	CBR Frequency	Member contacted 3+ times in 15 days	INFO
R-012	PMR No Follow-up	Open PMR with no notes past 5-day TAT	FLAG
R-013	Medical Records Via Email	Records submitted via email (not portal or fax)	INFO
R-014	Policy Change Impact	Claim type matches timing of recent policy change	INFO

Layer 2: Risk Scoring (Weighted Heuristic, 0–100)

Each claim is scored across seven feature categories:

Category	Weight	Example Features
Policy	15%	Policy age at claim, recent owner/beneficiary changes, coverage amount
Member	20%	Dependent count, dependent add velocity, prior claim count, claim frequency
Claim	20%	Amount relative to benefit max, documentation completeness, document format flags
Provider	15%	Total claims supported, unique patient count, approval rate, geographic concentration
Network	15%	Shared address count, shared provider with flagged members, family claim correlation
Temporal	10%	Proximity to termination, time between dependent add and claim, proximity to policy change
Rules Boost	5%	Number of rules triggered, maximum rule severity

Score interpretation:

Tier	Score Range	Queue Behavior
 HIGH	65–100	Priority review, top of queue
 MEDIUM	30–64	Examiner should investigate
 LOW	0–29	Likely clean, fast-track eligible

The risk breakdown is fully transparent — the examiner can see exactly which features contributed to the score and by how much.

Layer 3: Network/Link Analysis

An entity graph connects all members, dependents, providers, facilities, addresses, and employers. Pattern detection runs on this graph to find:

Pattern	Detection Method
Dependent Ring	Dependents connected to multiple unrelated members
Address Cluster	>3 unrelated members at the same address
Provider Mill	Provider with claim volume >2 standard deviations above peer mean
Employer Collusion	Claim spike from a single employer group
Cross-Member Dependent Sharing	Same dependent on multiple unrelated policies

Document Analysis Framework

Every claim's supporting documentation is evaluated against 13 checks. In the prototype, analysis is performed on document metadata attributes. In production, this integrates with document forensics services (OCR, image analysis, signature verification).

Check	What It Detects	Severity
DOC-001	Font inconsistency — multiple font styles in same document	High
DOC-002	Erasure indicators — evidence of erased or whited-out content	High
DOC-003	Typed over handwritten — typed text overlaying handwritten dates	High
DOC-004	Entirely handwritten report (unusual for modern facilities)	Medium
DOC-005	Missing standard headers (OVN, HCF)	Medium
DOC-006	Black & white records (possible altered photocopy)	Low
DOC-007	Editable Word document format (instead of PDF/image)	High
DOC-008	Inconsistent font sizes within a uniform document	Medium
DOC-009	Medical terminology misspellings (suggests non-medical author)	Medium
DOC-010	Suspicious or stamped-looking signature	High
DOC-011	Records submitted via email (not portal or fax)	Low
DOC-012	Missing vitals or medication list in full medical records	Medium
DOC-013	Records from a different provider than the requesting provider	Low

7-Step Investigation Checklist

A structured framework that can be run automatically or step by step. Each step has auto-pass criteria — if all criteria are met, the step passes without examiner intervention.

Step	Name	What It Checks	Auto-Pass Criteria
1	Initial Claim Review	Claim form completeness, submission channel, format	Required fields present, standard format
2	Eligibility Verification	Policy status, member identity, coverage confirmation	Policy active, member verified, benefit covered
3	Fraud Screening	Rules engine results, risk score, document checks	No rules triggered, risk <30, no document flags
4	Family & Pattern Review	Family claims (100 days), dependent legitimacy, cross-member patterns	Family claims in normal range, no dependent flags
5	Medical Record Review	Full document analysis, provider legitimacy, diagnosis match	All DOC checks pass, provider legitimate, diagnosis matches
6	Policy Terms Application	State rules, ET resolution, exclusions, policy alerts, coverage match	No exclusions apply, claim matches coverage
7	Determination	Approve / Deny / Pend / Escalate with documented rationale	Always requires human decision

The checklist serves two purposes:

1. **Automation** — tools run, results evaluated, clean claims fast-tracked
2. **Audit trail** — every step documented with findings, tools used, and outcome

What “Run Full Checklist” Returns

When the examiner clicks **Run Full Checklist** on a flagged claim (e.g., WC-247 — 35 dependents), the Copilot runs all 7 steps and returns output like this:

INVESTIGATION CHECKLIST – WC-247 (M. Rivera)		Risk: ● 94
● Step 1 – Initial Review ✓ All required fields present ✓ INITIAL_REVIEW task assigned	AUTO-PASSED	
● Step 2 – Eligibility Verification ✓ Policy is active ✓ Coverage type matches: wellness	AUTO-PASSED	
● Step 3 – Fraud Screening ❌ BLOCK rules triggered: R-001, R-002 R-001: Member has 35 dependents (threshold: 10) R-002: 47 claims filed across member + dependents in 90 days (threshold: 5 in 30 days) Risk score: 94.0 (HIGH) → Auto-adjudication blocked. Examiner review required.	FAILED	
● Step 4 – Family & Network Check 🚩 R-015: 35 family members filed claims in last 100 days 🚩 R-001: Dependent count (35) exceeds threshold (10) ✓ Shared address: Normal	NEEDS REVIEW	
● Step 5 – Medical Documentation Review ⚠️ DOC-001: Font inconsistency detected in 3 documents ⚠️ DOC-005: Missing standard medical record headers 🕒 Medical record request pending – waiting 4 days	NEEDS REVIEW	
● Step 6 – Policy & Coverage Determination ✓ Coverage matches: wellness ⚠️ PA-001: Policy alert active – surgical repair change does not apply to this claim type (wellness) ✓ No adverse state exclusions found	NEEDS REVIEW	
🟪 Step 7 – Final Determination Options: Approve / Deny / Pend / Escalate to SIU The above findings have been captured for your records.	AWAITING YOU	
Summary: 2 passed · 3 needs review · 1 failed		
Copilot: Steps 1–2 auto-passed. Step 3 failed – two BLOCK rules triggered (R-001, R-002). Recommend escalating to SIU given the 35-dependent pattern and claim velocity. Would you like me to compile the escalation dossier?		
[Compile Dossier] [Escalate to SIU] [View Network]		

Step status key:

- **PASS / AUTO-PASSED** — All criteria met; no examiner action needed
- **NEEDS REVIEW** — Flags or open questions; examiner should review findings
- **FAILED** — BLOCK rule triggered or critical failure; must be addressed
- 🟪 **AWAITING YOU** — Step 7 always requires a human decision

Banner branching (Step 3):

If the member has a `suspicious_banner` active in Prudential 360, Step 3 overrides all scoring and routes directly to senior examiner review — regardless of risk score. This mirrors the existing banner-to-review workflow, but now every claim still passes through Steps 1–2 and 4–6 automatically.

For a clean claim (risk score < 30, no rules triggered), the output looks like:

● Step 1 – Initial Review

● Step 2 – Eligibility Verification

● Step 3 – Fraud Screening

● Step 4 – Family & Network Check

● Step 5 – Medical Documentation

● Step 6 – Policy & Coverage

■ Step 7 – Final Determination

AUTO-PASSED

AUTO-PASSED

AUTO-PASSED (score: 12, LOW)

AUTO-PASSED

AUTO-PASSED (all 13 checks passed)

AUTO-PASSED

→ Fast-track eligible

Summary: 6 passed · 0 needs review · 0 failed

In this case the Copilot surfaces the result and recommends fast-track approval — the examiner confirms in one click.

Copilot Tool Set

The Copilot has access to 16 purpose-built tools organized into four categories:

Fraud Investigation Tools

Tool	Purpose	Key Output
check_eligibility	One-check eligibility verification (replaces 3 systems)	Policy status, member verification, coverage confirmation, issues found
check_family_claims	All claims from member + dependents in last 100 days	Claim list, total amounts, pattern analysis, comparison to normal range
analyze_medical_documents	Run all 13 DOC checks on claim documentation	Pass/fail per check, high-severity flags, plain-language summary
detect_dependent_anomalies	Dependent add patterns, age gaps, cross-member overlap	Dependent list, anomalies detected, add velocity, cross-member overlaps
check_provider_patterns	Provider claim volume, patient patterns, documentation quality	Monthly volume, unique patients, approval rate, peer comparison
flag_inconsistencies	Cross-reference all data for contradictions	Inconsistencies found plus innocent explanations for each
find_related_claims	Claims linked by provider, address, dependent, or pattern	Related claims by connection type, network risk level

Workflow Tools

Tool	Purpose	Key Output
check_workflow_tasks	Open PMR, CBR, and TAT status	Task details, days open, TAT compliance, action items
get_contact_history	Member contact count and history	Contact timeline, CBR threshold check, frustration risk

Policy Intelligence Tools

Tool	Purpose	Key Output
<code>get_policy_details</code>	Policy terms, age, owner/beneficiary history	Full policy record, change history, risk flags
<code>check_state_rules</code>	Applicable state rules and ET resolution	Governing state, ET hierarchy, exclusions, explanation
<code>check_policy_alerts</code>	Active policy changes affecting this claim	Applicable alerts, timing analysis, exploitation flag
<code>match_claim_to_coverage</code>	Compare claim details against policy terms	Coverage match, term-by-term comparison, amount determination

Universal Tools

Tool	Purpose	Key Output
<code>explain_risk_score</code>	Full risk score breakdown with feature contributions	Score, tier, top factors, all feature weights, rules triggered
<code>run_fraud_checklist</code>	Execute investigation checklist (all 7 steps or single step)	Step results, pass/fail/review status, next recommended step
<code>compile_dossier</code>	Generate investigation summary or escalation package	Timeline, evidence, network diagram, dollar exposure, recommendation

Prototype Dataset

The prototype operates on a fully synthetic dataset designed to demonstrate all platform capabilities:

Entity	Count	Notes
Employers	10	Mix of large, mid-size, and small
Members	200	Distributed across employers
Dependents	300	Realistic distribution + one outlier with 35
Providers	40	Physicians, hospitals, clinics
Facilities	15	Hospitals and surgical centers
Addresses	180	Most unique, some shared for network detection
Policies	200	Varying coverage, with change histories
Claims	500	Distribution below

Claim distribution:

Type	Count	Typical Behavior
Wellness	250	Most auto-adjudicated, high volume
Hospital Indemnity	100	Moderate complexity
Accident	100	Requires coverage matching
Critical Illness	50	High dollar, most scrutiny

Embedded Fraud Scenarios (10)

#	Scenario	Claim Type	Target Risk	Key Signals
1	35 Dependents Added	Wellness	94	Dependent spike, claim velocity, benefit max gaming
2	Termination Rush	Wellness + Accident	85	8 claims in final 30 days of coverage
3	Owner Change Before Critical Illness	Critical Illness	88	Policy owner changed 45 days before \$50K claim
4	Fabricated Accident	Accident	79	Injury mechanism inconsistent with medical records
5	Provider Mill	Hospital Indemnity	82	15 near-identical claims through one provider
6	Tampered Medical Records	Accident	76	Mixed fonts, erasures, typed over handwritten, Word doc
7	Duplicate Resubmission	Accident	71	Denied claim resubmitted with date changed by one day
8	Dependent Ring	Wellness	87	3 members from different employers share 8 dependents
9	Surgical Repair Exploit	Hospital Indemnity	65	4 claims filed day after outpatient surgery policy change
10	Missing Headers / B&W Records	Critical Illness	68	No standard headers, B&W photocopies, no vitals

Embedded False Positives (5)

These scenarios trigger flags but represent legitimate claims — demonstrating that the platform helps examiners avoid false accusations:

#	Scenario	Flag Triggered	Why It's Legitimate
A	New Employee Genuine Accident	R-003 (Early Filing)	Genuine accident, unfortunate timing — 60 days after policy start
B	Complex Claim Multiple Contacts	R-011 (CBR Frequency)	Complex claim, member genuinely needs updates
C	Rural Clinic B&W Records	DOC-006 (B&W Records)	Small rural clinic with old scanner
D	Blended Family Dependents	R-001 (Dependent Spike)	Marriage + stepchildren, 6 dependents added in 30 days
E	Genuine High Hospital Bill	R-009 (Benefit Max)	Hospital bill genuinely exceeded benefit maximum

Policy Intelligence

The platform tracks active policy changes and proactively surfaces them when relevant:

Current Active Alert: PA-001

Field	Detail
Effective Date	September 18, 2025
Change	Surgical repair benefits now paid regardless of hospital confinement
Previous Rule	Surgical repair only when confined to hospital
New Rule	Surgical repair paid for hospital or outpatient
Applies To	Hospital Indemnity claims
Platform Behavior	Proactively alerts examiner when reviewing affected claims; flags claims filed suspiciously close to the effective date

Extra-Territorial (ET) Rules

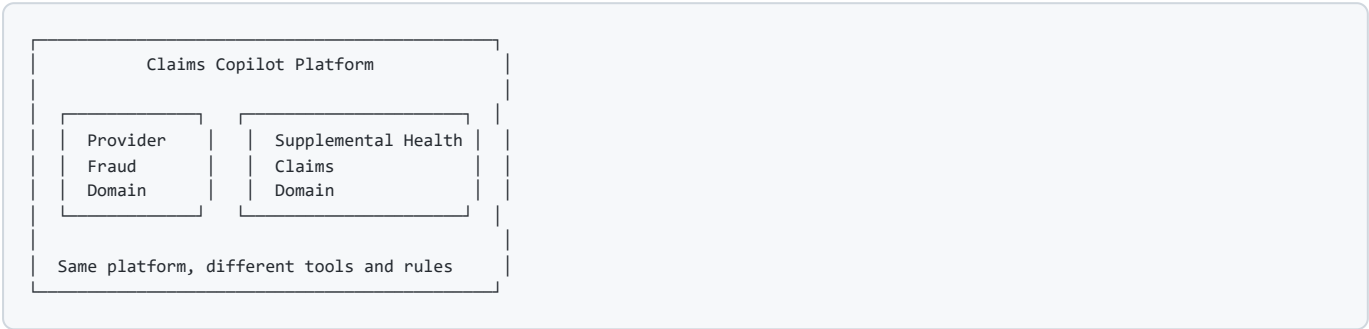
The platform resolves ET hierarchy automatically:

- Most recent ET supersedes the policy's governing state
 - State-specific exclusions are surfaced (e.g., Florida intoxication exclusion)
 - Plain-language explanation of which rules apply and why
-

Architecture & Design Principles

Domain-Agnostic Platform

The platform is built on a configurable architecture. The supplemental health claims domain is one configuration — the same platform supports provider fraud investigation (included in the prototype) and can be extended to additional lines of business by registering a new domain configuration.



Examiner-First Design

The platform is designed as a **daily-use workflow tool**, not a fraud alert system. The distinction matters:

- **Fraud alert system:** Examiners use it when something looks suspicious
- **Workflow platform:** Examiners use it on every claim; fraud detection is embedded

This approach drives adoption because the platform provides value on routine claims (faster eligibility checks, automated policy matching, workflow task tracking) — not just on the rare fraud case.

Transparent AI

The Copilot never makes final determinations. It presents evidence, explains risk scores with full feature breakdowns, and always includes innocent explanations alongside red flags. The examiner decides.

Prototype Scope

Component	Prototype	Production
Claims data	500 synthetic claims	Live claims feed
Document analysis	Metadata-based simulation	OCR, image forensics, signature verification
Risk scoring	Weighted heuristic	ML model trained on examiner confirm/dismiss actions
Entity graph	Pre-computed at startup	Real-time graph database
Policy rules	Single active alert (PA-001)	Full policy change feed
State rules	Simulated ET resolution	Complete state rule engine

Technology Stack

Layer	Technology
Frontend	React + Vite
API	FastAPI (Python)
AI Agent	LangGraph + Amazon Bedrock (Claude)
Graph	NetworkX
Communication	WebSocket (real-time chat)
Data	Synthetic generation, in-memory cache

Demonstration Walkthrough

The prototype supports a guided 7-minute demonstration:

1. **The Morning View** (0:30) — Claims Queue with 500 scored and ranked claims
2. **Routine Claim** (1:30) — Process a low-risk accident claim in 2 minutes: eligibility, coverage matching, state rules, workflow tasks
3. **Policy Intelligence** (0:30) — Proactive surgical repair policy alert
4. **Flagged Claim** (0:30) — 35-dependent case: risk score breakdown, rules triggered
5. **Automated Checklist** (0:45) — 7 steps execute, step 4 goes red
6. **Document Fraud** (0:30) — Tampered records: 5 DOC flags identified
7. **Network Discovery** (0:30) — Dependent ring visualized across 3 members
8. **False Positive** (0:30) — “What am I missing?” — blended family is legitimate
9. **Escalation** (0:30) — Auto-generated SIU referral package
10. **Dashboard** (0:30) — \$12,400 held, patterns detected, workflow health
11. **Domain Agnostic** (0:30) — Switch to provider fraud domain, same platform

This document describes a working prototype. All data is synthetic. Document analysis operates on simulated metadata. Risk scoring uses a weighted heuristic; production deployment would incorporate machine learning trained on examiner decisions.